

From grading to growth

1 Problem

Large-enrolment modules struggle to provide meaningful feedback on open-ended assessments:

- ⌚ Manual grading can be slow & difficult to scale consistently.
- 💬 Students receive marks or brief annotations without thoroughly understanding their mistakes.
- 🔍 Instructors have limited visibility into cohort-wide misconceptions.
- 💡 Students lack practice targeted to their specific learning gaps.

WHY FEEDBACK MATTERS

94.92% of students said feedback after a quiz was very important to them — and the majority wanted it in a timely manner, not at term's end.
– Edokpayi, J. N. (2025). Assessing the impact of frequent quizzes on student performance. *Discover Education*, 4(480).

2 Our Solution

AIMS is a human-in-the-loop AI platform that:

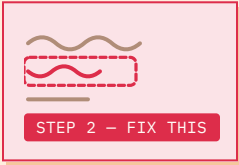
- 👁️ Capable of taking both handwritten and typed/digital responses as input.
- 📋 Suggests rubric-aligned scores with confidence indicators.
- 🎯 Detects misconceptions and valid alternative approaches.
- 💬 Generates personalised feedback and targeted practice.
- 👤 Keeps instructors in control of all grading decisions.

3 Impact

Streamlined, accelerated marking: AIMS takes on the cognitive work of reading each response and mapping it to the rubric, so the instructor's time goes to verifying and assigning the mark — not decoding the handwriting first. The instructor still makes the final decision on every submission.

Personalised feedback at scale: explains what was correct, incomplete, or misunderstood.

Improved mastery: transforms weekly quizzes into a continuous learning cycle.



WHAT STUDENTS GET WITH AIMS

The exact step that went wrong, why it matters, and a practice question to fix it.

Existing platforms vs. AIMS

EXISTING GRADING PLATFORMS	AIMS (OUR UNIQUE SELLING POINTS)
Streamlines rubric-based marking	Supports marking for instructors AND targeted learning for students
Feedback depends largely on manual comments	Automatically drafts misconception-specific feedback
Provides general assignment results	Identifies personalised individual & cohort learning gaps and provides targeted practice questions
Ends after grades are released	Connects mistakes to targeted practice
Rubrics are adjusted as the instructor is marking submissions	Suggests rubric refinements to the instructor before detailed submission review

One engine, two release paths

Every submission takes the same first two steps. What happens next depends on what's at stake.

TOOLS USED pix2text AWS Textract OpenAI API 5.6 RAG SymPy

Proposed solution: AIMS digitises the script, drafts a suggested rubric-matched score, and names the misconception. It closes the loop with feedback and targeted practice, and the instructor verifies and finalises every mark before release.

INSTRUCTOR DASHBOARD pictured: the summative review screen

SAMPLE QUESTION · 1 OF 3

SUMMATIVE

ORIGINAL SCRIPT (SCANNED)

Sequence $u_1, u_2, u_3, \dots$ where $u_{n+1} = 2u_n + An$, $u_1 = 5$, $u_2 = 15$. Find A , u_3 ; find a, b, c ; find $\sum u_r$.

Part (i)

$15 = 2(5) + A$
 $A = 5$

Part (ii)

$a = 7.5$, $b = -5$, $c = -5$

Part (iii)

$\sum u_r = 15(2^{n+1} - 1) - 5n(n+1)/2 - 5n$

RECONCILED STEPS (one representative step per part)

STEP 1 PART (I) CONF 95%

$15 = 2(5) + A$, $A = 5$
Solves for A using the recurrence relation.
[view LaTeX](#)

STEP 2 PART (II) CONF 84%

$a = 7.5$, $b = -5$, $c = -5$
Solves the system of equations for a, b, c.
[view LaTeX](#)

STEP 3 PART (III) CONF 81%

$\sum u_r = 15(2^{n+1} - 1) - 5n(n+1)/2 - 5n$
Finds the sum of the sequence in terms of n.
[view LaTeX](#)

AI Summary

Recommends 7/10. Strongest on "Part (ii) - Finding a, b, and c", weakest on "Part (iii) - Finding the sum of u_r ".
[edit](#)

RUBRIC (RAG-MATCHED)

Part (i) - Finding A and u_3

RAG

+1

Correctly finds $A=5$ but does not calculate u_3 .

Step 1 novice · 1 proficient · 2

Part (ii) - Finding a, b, and c

RAG

+4

Forms all three equations, solves for $a=15/2$, $b=-5$, $c=-5$.

Step 2 novice · 2 proficient · 4

Part (iii) - Finding $\sum u_r$

RAG

+2

Sets up the summation but misapplies the geometric series step.

Step 3 novice · 2 proficient · 4

Total

7 / 10

Approve & next

Edit step

Skip

▶ Watch the demo: AIMS in action

[YouTube link]

